NOAA Technical Memorandum NMFS-XXX-##

Caribbean Ecosystem Status Report

Gulf of Mexico Integrated Ecosystem Assessment Program

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U.S. DEPARTMENT OF COMMERCE

National Oceanic and Atmospheric Administration
National Marine Fisheries Service
Northwest Fisheries Science Center

Caribbean Ecosystem Status Report

Gulf of Mexico Integrated Ecosystem Assessment Program[1]

1. NOAA Fisheres, Southeast Fisheries Science Center

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Executive Summary

TEST making a change

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1 Introduction

1.1 About this report

This report is for the Caribbean Fishery Management Council (CFMC). The purpose of this report is to synthesize ecosystem and socioeconomic information to help the CFMC better meet fishery management objectives. This report relied on both previously identified proposed indicators as well as expert vetting to select a suite of indicators that best address the fishery management plan (FMP) objectives for the U.S. Caribbean. Information in this report is organized into two sections: tracking performance toward fishery management objectives and potential risks to meeting fishery management objectives. The style of this report is based on the 2023 State of the Ecosystem Reports for the Northeast U.S. Shelf.

1.2 Report structure

The CFMC's Science and Statistical Committee, as well as the region's Ecosystem-Based Fishery Management Technical Advisory Panel (EBFM TAP), recently completed a series of conceptual models linking key components of the ecosystem and human activities related to fishing. This report used these conceptual models as a starting list of proposed indicators and matched the indicators to answer FMP objectives when possible. For those objectives that did not have an immediate conceptual model-identified indicator, this report used a decision matrix process for expert vetting (Figure 1.1).

This decision matrix was composed of a list of proposed indicators compiled from the conceptual models as well as proposed indicators provided via expert input. These potential indicators were vetted and edited by expert small working groups, who then scored a decision matrix (Figure 1.1) of potential indicators against the following decision criteria: long term data availability, measurability, sensitivity to environmental changes, specificity, spatial and temporal scalability, relevance to specific FMP objectives, and responsiveness to management actions.

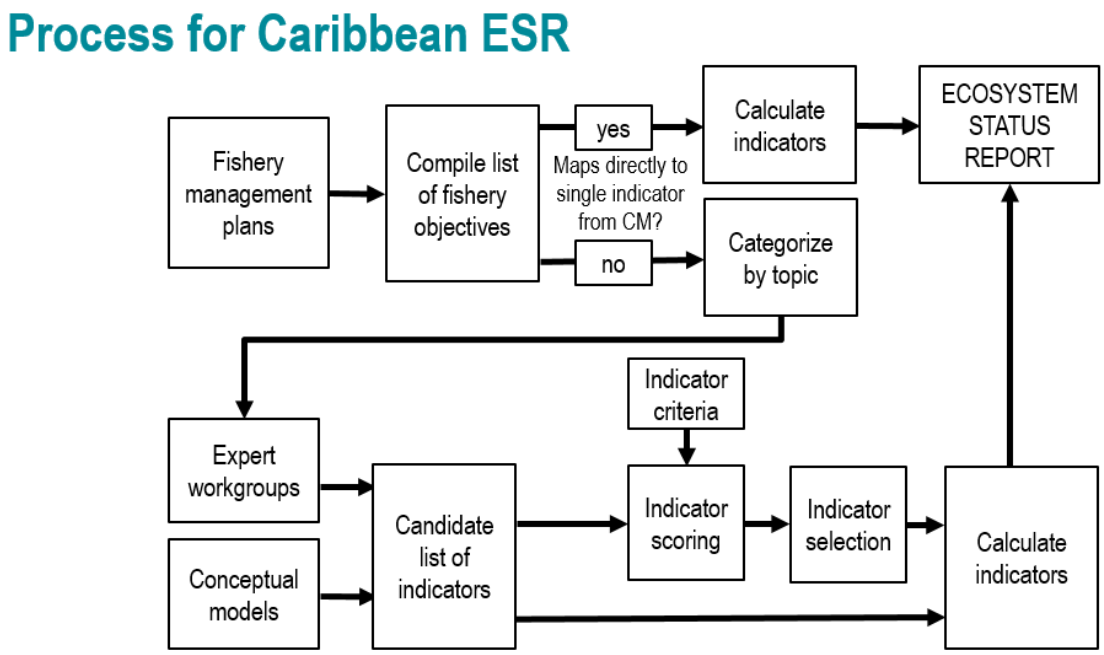


Figure 1.1: Process for selecting indicators for the U.S. Caribbean Ecosystem Status Report.

1.3 Notes on interpreting time series figures

Time series data are plotted in a standardized format for ease of interpretation (e.g., Figure 1.2). The x-axis represents the temporal dimension, which may be monthly, yearly, or irregular time steps, and the y-axis represents the indicator value in units specified in the axis label. The dashed horizontal line represents the mean indicator value across the entire time series, and the solid horizontal lines denote the mean plus or minus one standard deviation. Red shaded areas and green shaded areas show years for which the indicator value is below or above one standard deviation from the mean, respectively. The blue vertical shaded box highlights the last five years of indicator values, over which additional metrics are calculated. Black circles to the right of each figure indicate whether the indicator values over the last five years are greater (plus sign), less than (minus sign), or within (solid circle) one standard deviation from the mean of the overall time series. Arrows to the right of each figure indicate whether the least squares linear fit through the last five years of data produces a positive or negative slope that is greater than one standard deviation (upward or downward arrows respectively), or less than one standard deviation (left-right arrow).

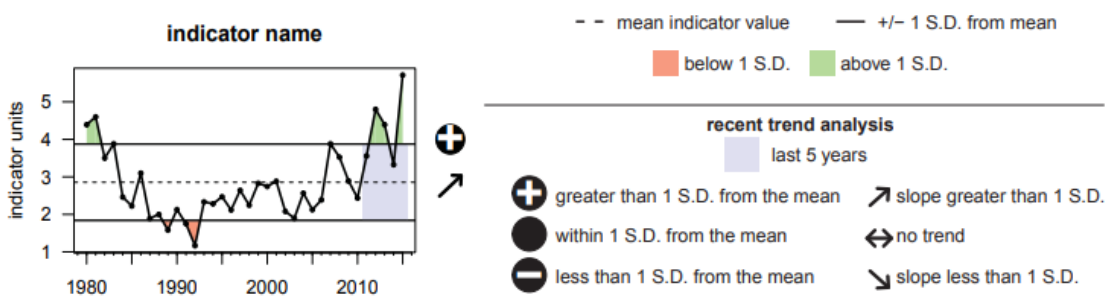


Figure 1.2: Example time series plot, showing an indicator plotted with its mean and standard deviation, and trend analysis for the most recent five years of data. See text for more detailed description of specific calculations.

2 Risks to meeting fishery management objectives

In this section, we examine indicators...

2.1 Sea surface temperature

Ocean temperatures affect species distributions and other aspects of population dynamics and have impacts on habitats such as coral reefs. Monthly mean, minimum, and maximum sea surface temperatures were calculated based on the 1/4 Degree Daily Optimum Interpolation Sea Surface Temperature (OISST) Analysis (cite). Mean temperatures in the U.S. Caribbean region have been increasing at an average rate of 0.25 degrees C per decade. In the last five years, minimum temperatures have been well above average (Fig), while there has been no long-term or recent trend in maximum temperatures experienced.

2.2 Coral bleaching stress

Accumulated heat stress, which can lead to coral bleaching and death, is measured by summing degree heating weeks for the most recent 12-week period from sea surface temperature data (cite). Bleaching stress was generally below average until the mid-2000s, until a sudden bleaching event in 2005 which is now the second most severe event in history. In 2024, and unprecedented bleaching event occurred across the U.S. Caribbean and beyond (Figure 2.2).

2.3 Ocean acidification via aragontite saturation state

Ocean and coastal acidification can impact organisms directly or indirectly; a decrease in aragonite saturation state can weaken the structure of coral reefs and other calci-

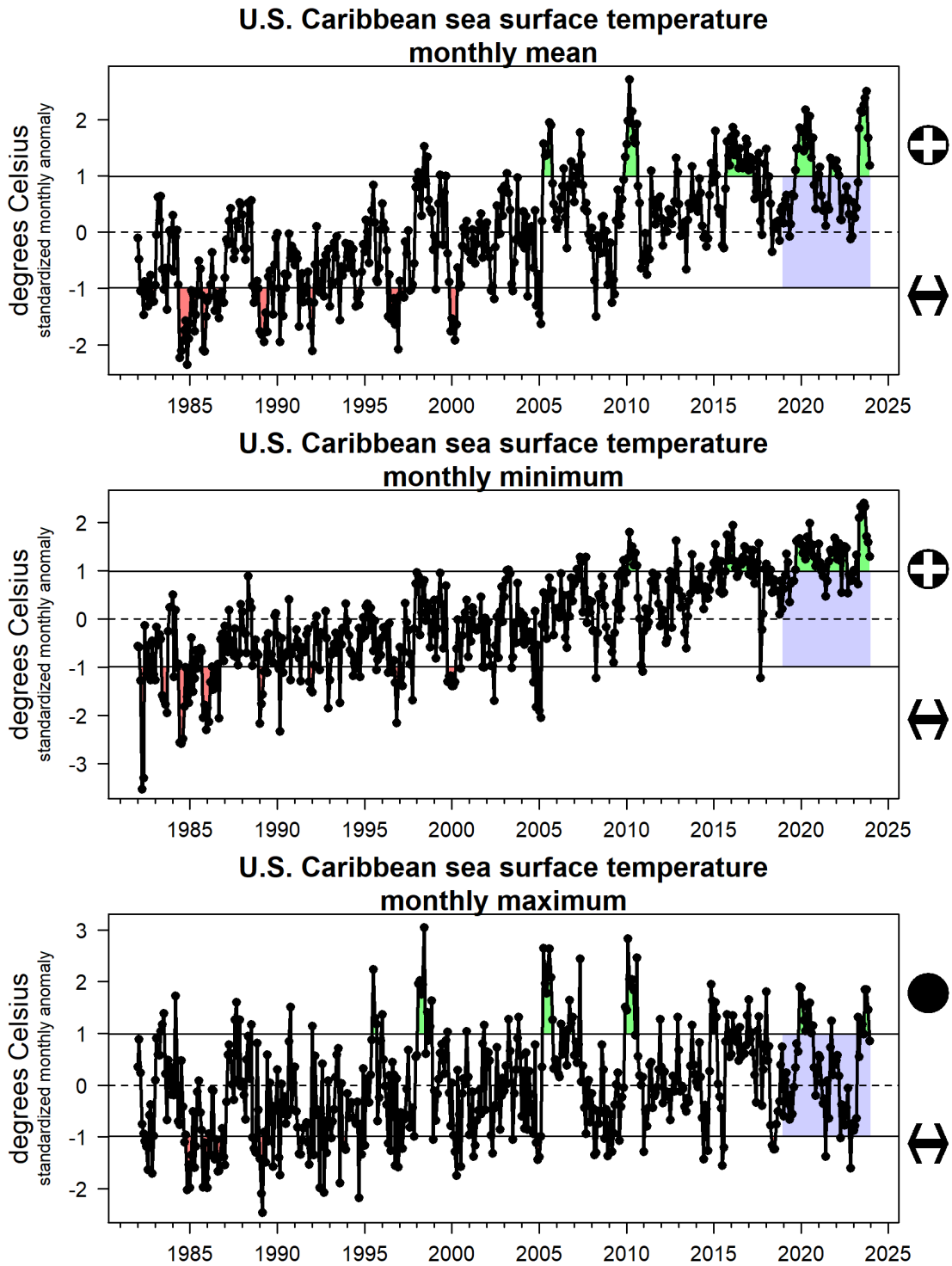


Figure 2.1: SST plot

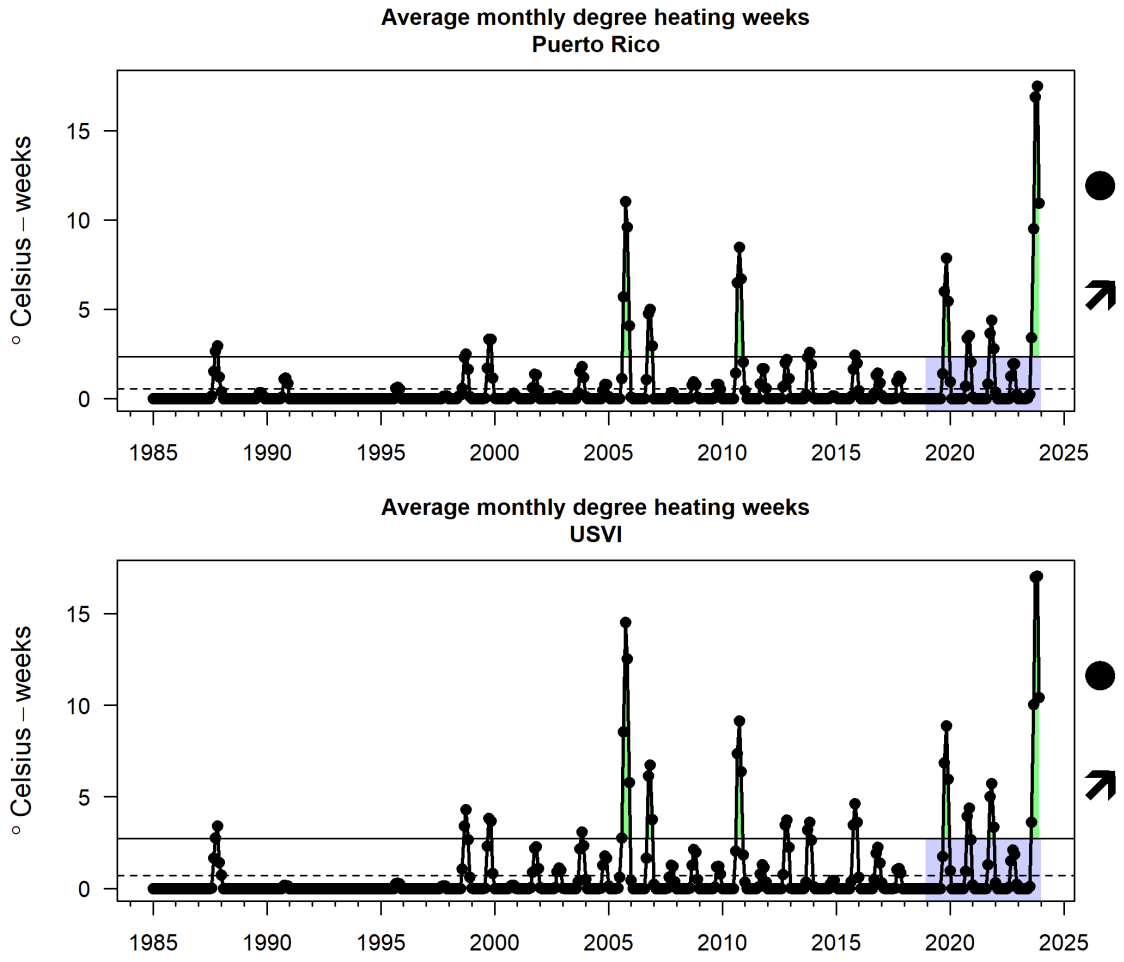


Figure 2.2: Number of degree heating weeks in Puerto Rico and the USVI as a measure of thermal stress to corals.

fying organisms. In-situ measurements of aragonite saturation states are scarce and a synoptic, long-term view is only available from modeled products. Aragonite saturation state was derived for the U.S. Caribbean region from the MOM-TOPAZ hindcast (cite). An overall negative trend occurs, with an acceleration of this trend apparent after 2008 (Fig).

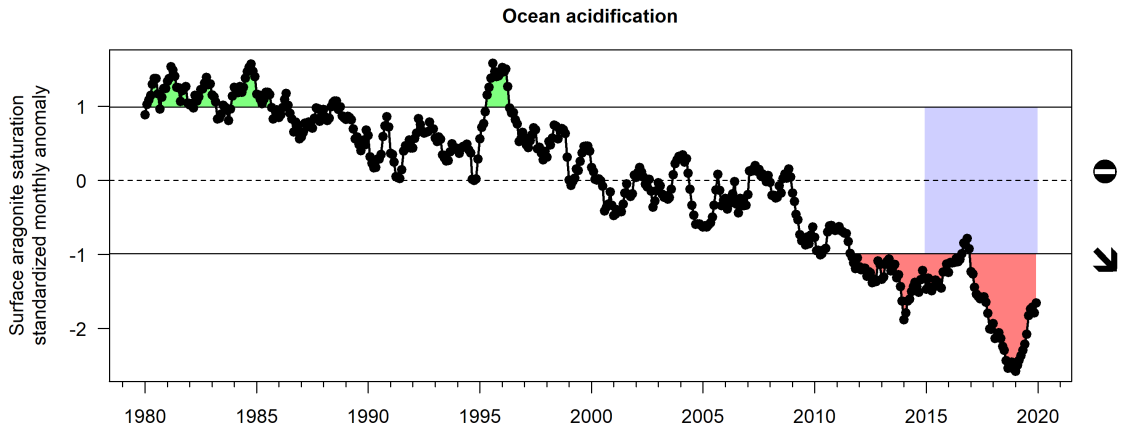


Figure 2.3: Ocean acidification plot

2.4 Hurricane activity

Hurricane activity can be captured by the accumulated cyclone energy index which is calculated as the sum of squared wind speeds for storms tracking through the U.S. Caribbean region, as documented by the International Best Track Archive for Climate Stewardship database (cite). The index has fluctuated throughout the past seven decades, with multiple notable peaks (Fig). The year 2017 hurricane activity was at an unprecedented high, due to two major hurricanes that struck the islands: Irma and Maria.

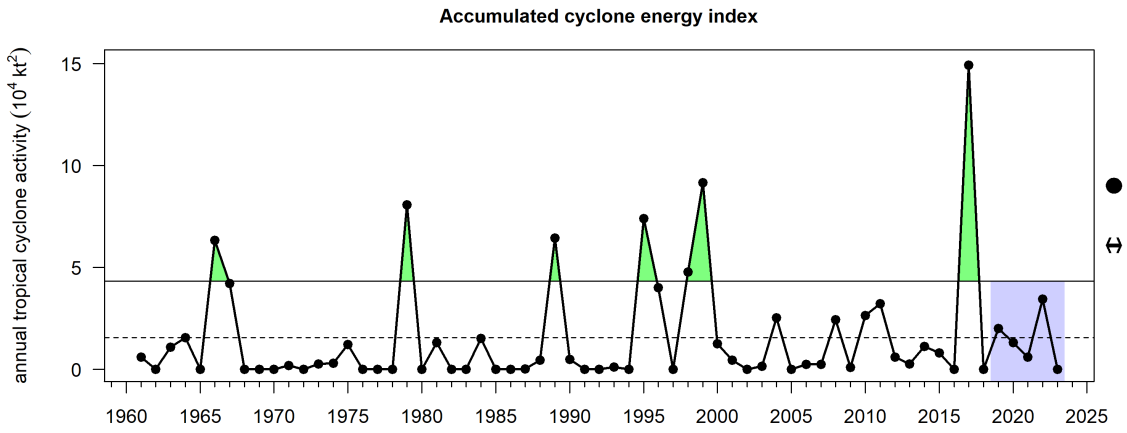


Figure 2.4: Hurricane activity

2.5 Number of major earthquakes

Earthquakes in Puerto Rico can induce landslides and cause impacts to infrastructure including homes and the electrical grid, and can be a source of stress in the affected human population (Agar et al. 2022 NOAA Technical Memorandum NMFS-SEFSC-759). Seismic events are reported by the USGS in near real-time (cite). A major earthquake swarm occurred in Southwest Puerto Rico in early 2020; in this year there were over 400 events of greater than 3.5 magnitude on the Richter scale (Fig).

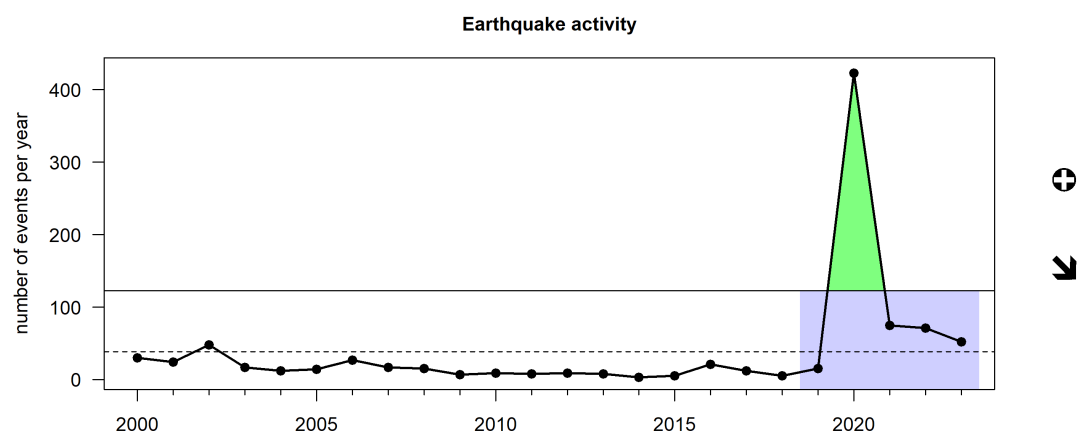


Figure 2.5: Earthquakes plot

2.6 Identified point source pollution sites

Impacts from terrestrial pollution can be captured from several databases maintained by the Environmental Protection Agency, which provide information on companies that have been issued permits to discharge wastewater into rivers, release of toxic chemicals and waste management activities at facilities, and declaration of Superfund sites. The number of pollution sites reported increased in the 2000s, but has decreased slightly in both Puerto Rico and USVI in recent years (Fig). Note that this indicator does not represent the timing of when pollution was impacting the ecosystem, but rather the timing of political action or attention on the environmental impacts of pollution.

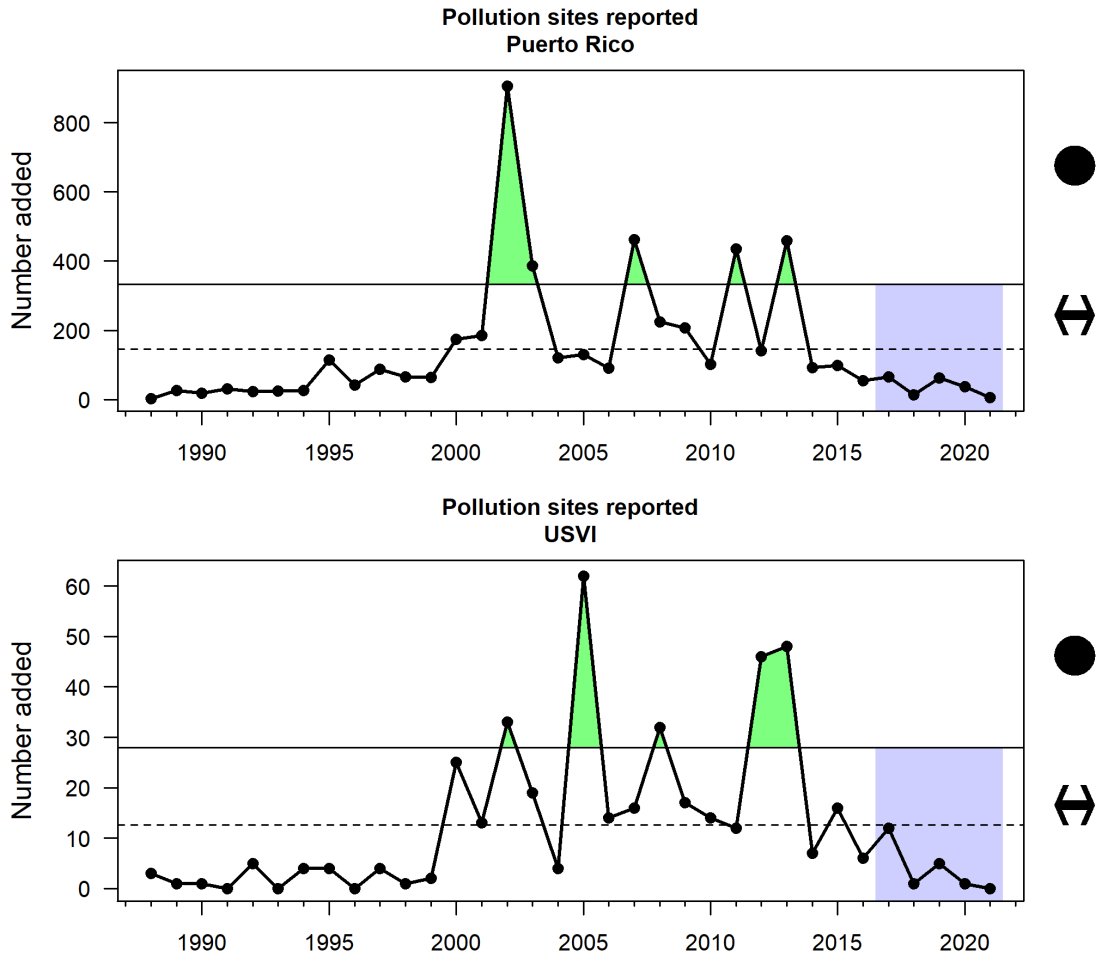


Figure 2.6: Pollution plot

2.7 Turbidity

Coastal pollution, runoff, and water quality issues are of major concern to fishing-dependent communities in the U.S. Caribbean (Seara et al. 2024). Water clarity can be measured by the diffuse attenuation coefficient which indicates how strongly light intensity is attenuated within the water column; however, satellite sensors cannot differentiate between organic and inorganic water particles contributing to water clarity. NOAA's Coastwatch program provides estimates of the attenuation coefficient for

penetration of light at 490nm (Wang et al. 2009) based on multiple satellite sensors. [sentence on trends]

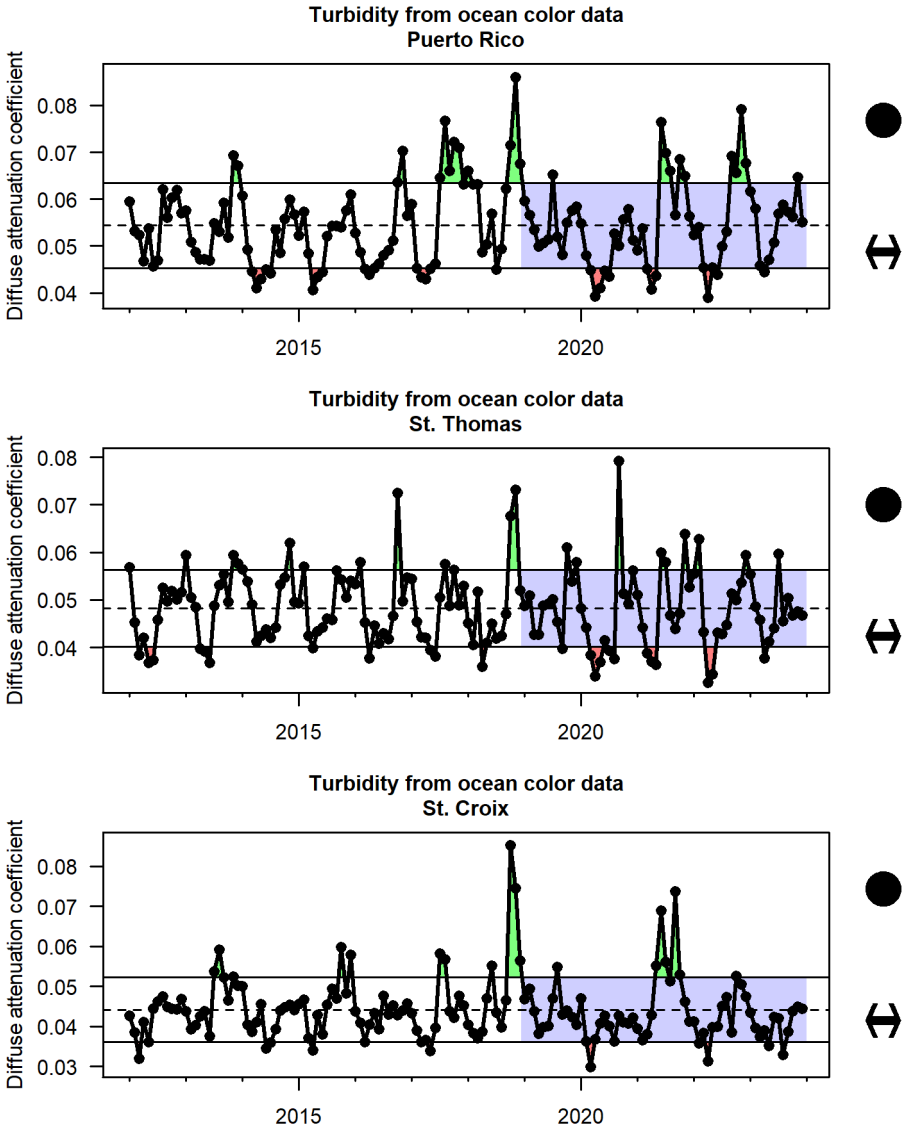


Figure 2.7: Turbidity plot

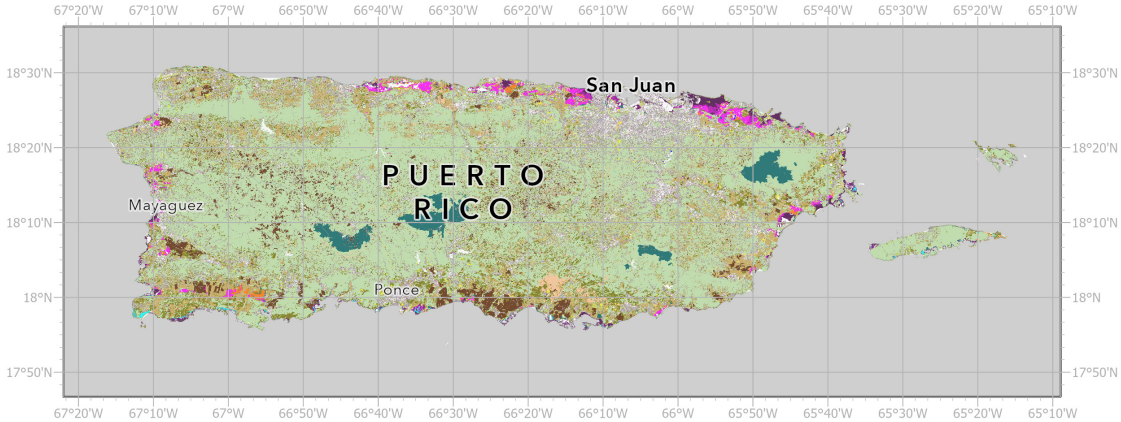
2.8 Water quality

WORKING ON THIS ONE

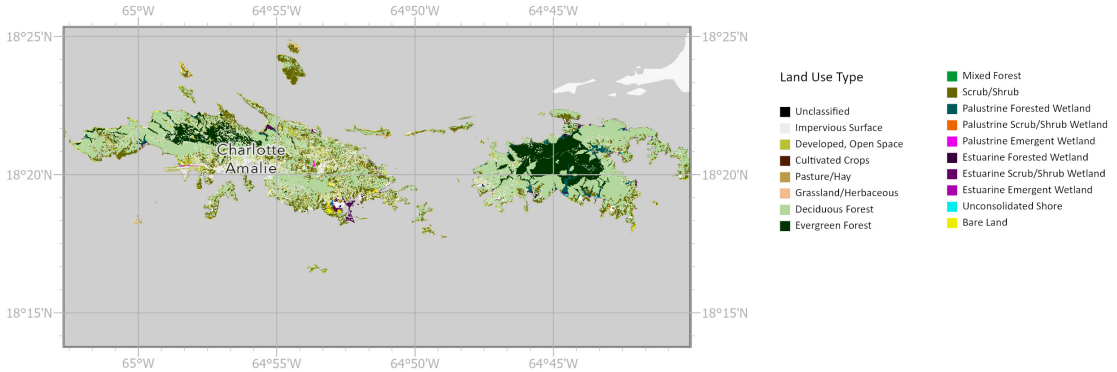
2.9 Coastal development

Indicator 10

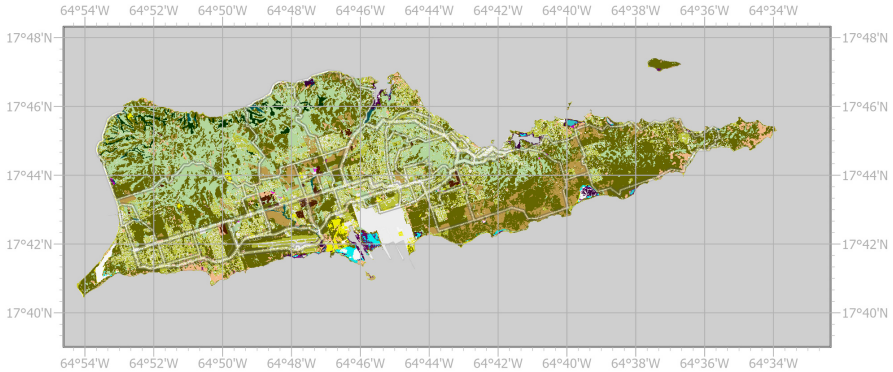
Puerto Rico



St. Thomas and St John



St. Croix



2.10 Primary productivty via ocean color

Primary productivity is a measure of the total energy available in an ecosystem and is closely correlated with chlorophyll a concentrations. Average chlorophyll a concentrations are derived from the European Space Agency Climate Change Initiative's Ocean Colour product which provides a bias-corrected composite of measurements merged from multiple satellite sensors (cite). Concentrations are plotted as standardized monthly anomalies as there is a seasonal signal that could mask long-term trends. Estimates show a decadal cyclical pattern, with no overall or recent trend apparent (Fig).

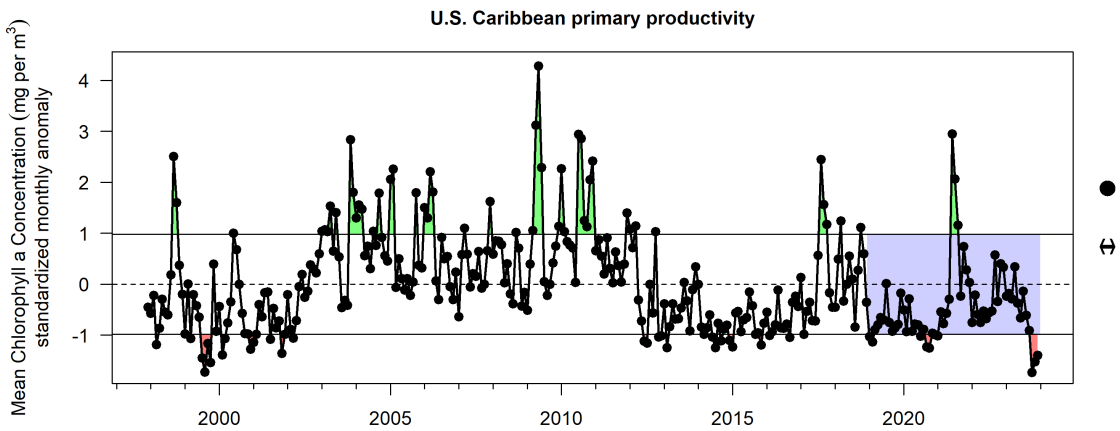


Figure 2.8: PP plot

2.11 Sargassum innundation

Sargassum (brown macroalgae S. Fluitains and S. natans) is a designated essential fish habitat important for many pelagic fish and protected species; however, when large blooms collect in nearshore environments the mats can reduce oxygen, suffocate beaches and have detrimental impacts on marine species. Mean monthly Sargassum wet biomass is estimated from satellite measurements using the algorithm of Wang et al. (2018). Sargassum blooms were largely absent from the U.S. Caribbean prior to 2011, but bloom activity has been generally increasing since that year (Fig). Major inundation events occurred in 2018 and 2021.

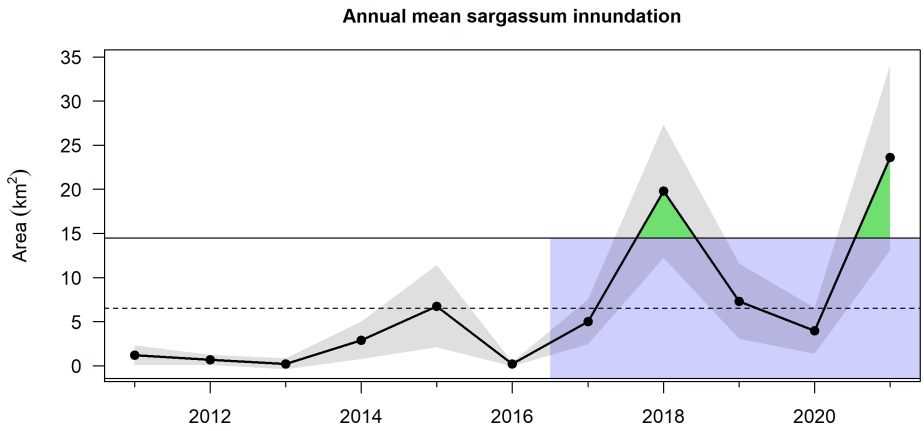


Figure 2.9: Sargassum plot

2.12 Market disturbances

Alterations to typical fishing patterns can be quantified by analyzing the seasonality of how fishing activity is distributed throughout the year and detecting deviations from average patterns. A market disturbance indicator was developed by calculating the proportion of landings in each month of the year, and summing the square of deviations between those monthly proportions from the mean proportions across all years. In Puerto Rico there is little trend in the indicator; however there were disturbances in 2005 and 2020-2021. In St. Thomas, the indicator increases throughout time and detects a major disturbance in the 2017-18 fishing season. In St. Croix, disturbance levels were high in 2017-18 and also 2019-20.

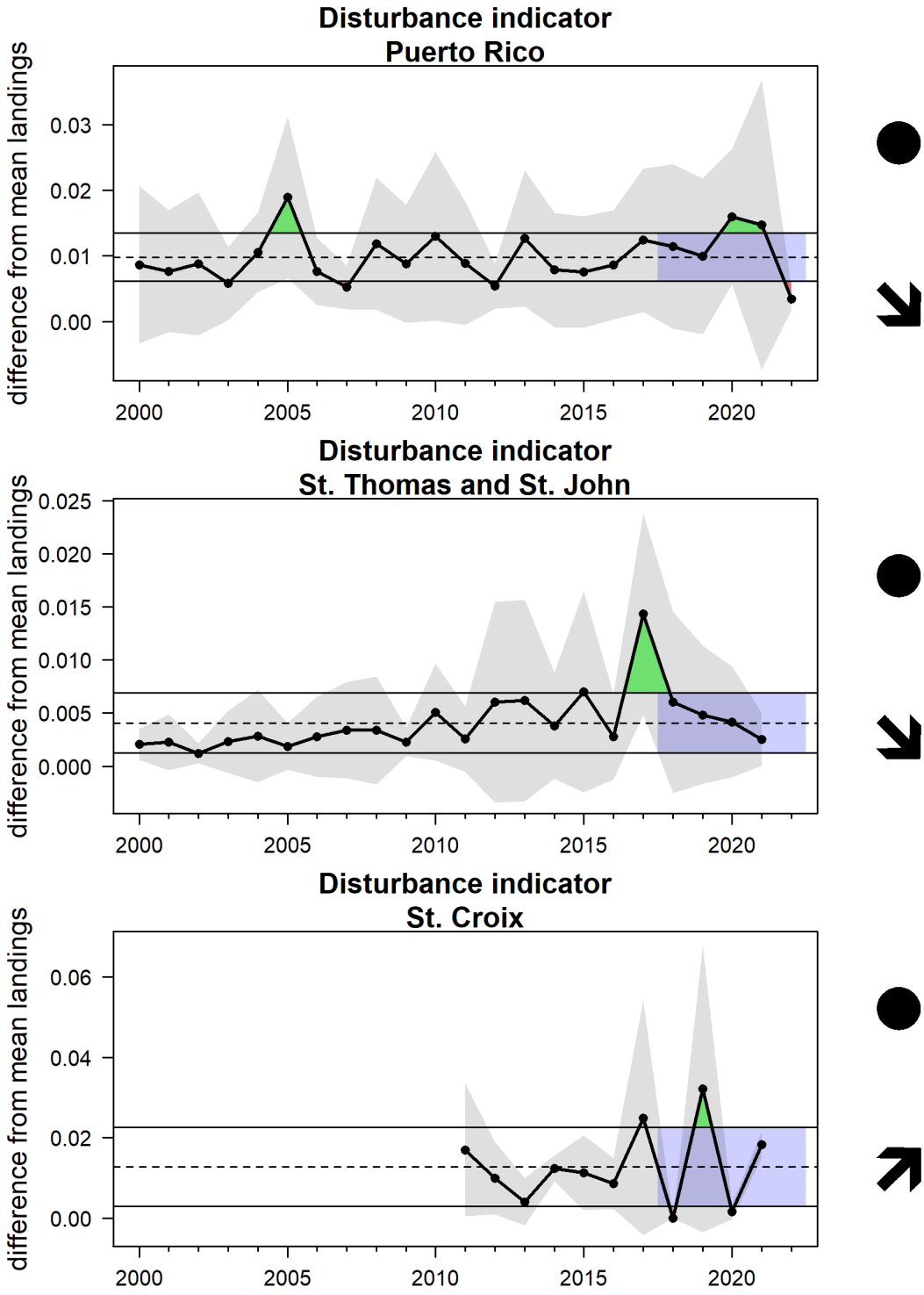
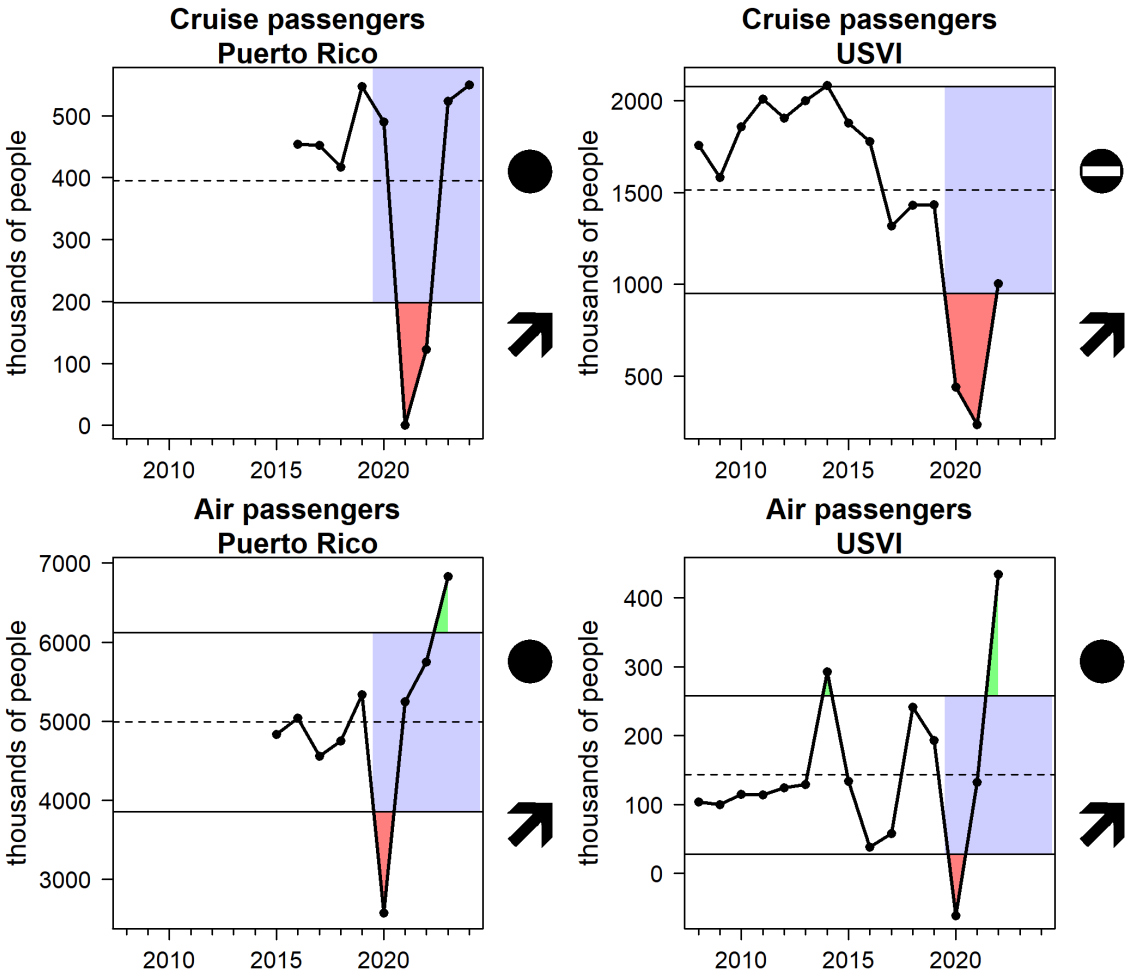


Figure 2.10: Disturbance plot

2.13 Tourism (air arrivals and cruise passengers)

Human activity has an impact on the marine ecosystem indirectly through its influence on coastal development and pollution, as well as directly through marine tourism, fishing and demand for seafood. Human activity is exerted by the local population as well as the extensive tourism industry that exists in the U.S. Caribbean. Total population estimates are reported by the U.S. Census (??) and tourism activity can be measured through hotel occupancy rates (cite) and the number of cruise passengers (cite). Human population in the U.S. Caribbean has been declining gradually since 2000. Tourism was relatively stable until recent years, with major decreases in 2017 and 2020.



Human population

Indicator 15

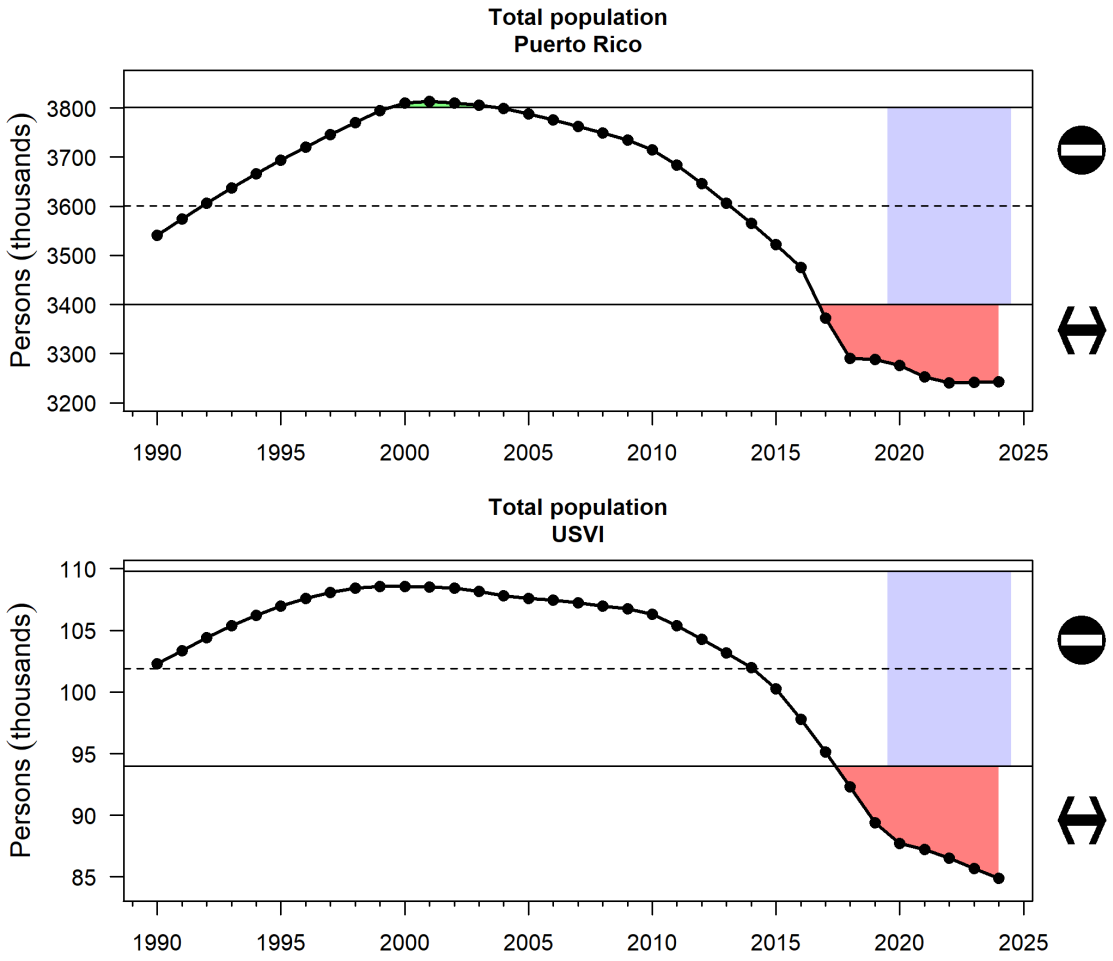


Figure 2.11: Population plot

3 Customization

3.1 Quarto documentation

Quarto allow many bells and whistles to make nice output. Read the documentation here Quarto documentation.

3.2 Examples

Looking at other people's Quarto code is a great way to figure out how to do stuff. Most will have a link to a GitHub repo where you can see the raw code. Look for a link to edit page or see source code. This will usually be on the right. Or look for the GitHub icon somewhere.

- Quarto gallery
- nmfs-openscapes
- Faye lab manual
- quarto-titlepages Note the link to edit is broken. Go to repo and look in `documentation` directory.

4 Rendering

The repo includes a GitHub Action that will render (build) the website automatically when you make changes to the files. It will be pushed to the `gh-pages` branch.

But when you are developing your content, you will want to render it locally.

4.1 Step 1. Make sure you have a recent RStudio

Have you updated RStudio since about August 2022? No? Then update to a newer version of RStudio. In general, you want to keep RStudio updated and it is required to have a recent version to use Quarto.

4.2 Step 2. Clone and create RStudio project

First, clone the repo onto your local computer. How? You can click File > New Project and then select "Version Control". Paste in the url of the repository. That will clone the repo on to your local computer. When you make changes, you will need to push those up.

4.3 Step 3. Render within RStudio

RStudio will recognize that this is a Quarto project by the presence of the `_quarto.yml` file and will see the "Build" tab. Click the "Render website" button to render to the `_site` folder.

Previewing: You can either click `index.html` in the `_site` folder and specify "preview in browser" or set up RStudio to preview to the viewer panel. To do the latter, go to Tools > Global Options > R Markdown. Then select "Show output preview in: Viewer panel".

5 Figures and Tables

Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

5.1 Code

You can embed an R code chunk like this:

```r
summary(cars)
```

```
     speed           dist       
 Min.   : 4.0   Min.   :  2.00  
 1st Qu.:12.0   1st Qu.: 26.00  
 Median :15.0   Median : 36.00  
 Mean   :15.4   Mean   : 42.98  
 3rd Qu.:19.0   3rd Qu.: 56.00  
 Max.   :25.0   Max.   :120.00  
```

5.2 Including Plots

You can also embed plots and reference them, like so Figure 5.1.

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

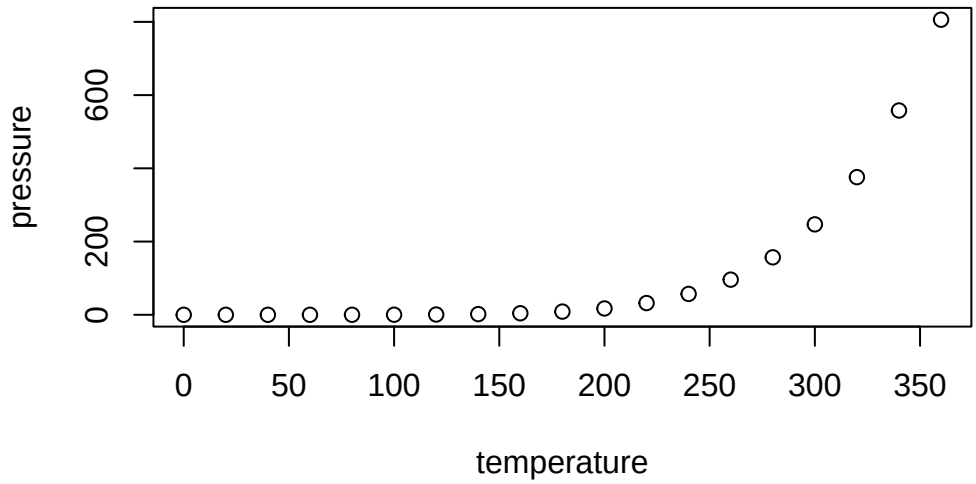


Figure 5.1: Plot of pressure

5.3 Including Tables

You can also embed tables and reference them with Table 5.1.

```
library(knitr)
kable(head(iris))
```

Table 5.1: Iris Data

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

6 Rendering with Code

You can have code (R, Python or Julia) in your qmd file. You will need to have these installed on your local computer, but presumably you do already if you are adding code to your qmd files.

```r
x <- c(5, 15, 25, 35, 45, 55)
y <- c(5, 20, 14, 32, 22, 38)
lm(x ~ y)
```

```
Call:
lm(formula = x ~ y)

Coefficients:
(Intercept)            y
      1.056        1.326
```

6.1 Modify the GitHub Action

You will need to change the GitHub Action in `.github/workflows` to install these and any needed packages in order for GitHub to be able to render your webpage. The GitHub Action install R since I used that in `code.qmd`. If you use Python or Julia instead, then you will need to update the GitHub Action to install those.

If getting the GitHub Action to work is too much hassle (and that definitely happens), you can alway render locally and publish to the `gh-pages` branch. If you do this, make sure to delete or rename the GitHub Action to something like

```
render-and-publish.old_yml
```

so GitHub does not keep trying to run it. Nothing bad will happen if you don't do this, but if you are not using the action (because it keeps failing), then you don't need GitHub to run it.

6.2 Render locally and publish to gh-pages branch

To render locally and push up to the `gh-pages` branch, open a terminal window and then `cd` to the directory with the Quarto project. Type this in the terminal:

```
quarto render gh-pages
```

7 References

Quarto has powerful references functionality. You can easily insert citations from Zotero libraries that you maintain in the cloud (on Zotero). This allows the whole team to update the library and you can sync up to that library. Read about this on the Quarto documentation on citations. Google youtube videos on this also to see it in action.

Add a `.bib` file in to your project or add a linked Zotero library via RStudio in Visual mode with Tools > Project Options... > R Markdown > select custom libraries from the Zotero dropdown.

The you can type `@` and you will see a dropdown of the references in your libraries. You can then select the ones to add. If you don't see the one you need, you can paste in the DOI and it will be added to your references file (with all the info). The references will be added to your references section of your book automatically.

See the `references.qmd` file for how to include the references.

- `@ansley1981` will produce Ansley and Davis (1981)
- `[@ansley1981]` will produce (Ansley and Davis 1981).

References

Ansley, H. L. H., and C. D. Davis. 1981. "Migration and Standing Stock of Fishes Associated with Artificial and Natural Reefs on Georgia's Outer Continental Shelf." Brunswick, Georgia, USA.